Real-Time Dense Map Matching with Naive Hidden Markov Models

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UE24CS352A Machine Learning | Section: Not provided | Problem: Not provided

Problem

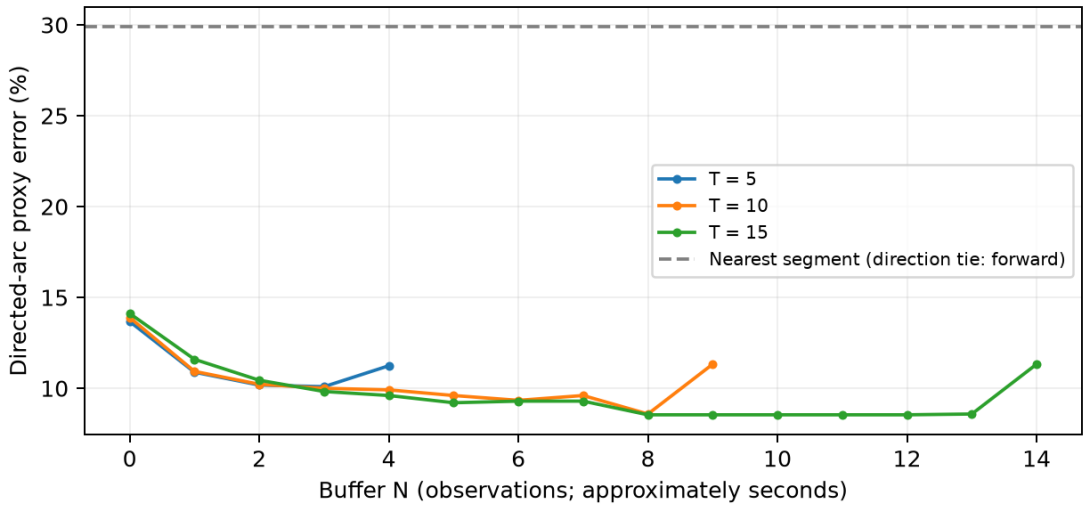
Assign noisy vehicle GPS observations to directed road arcs in real time, without shortest-path searches. The objective is to implement the naive HMM approach of Ahres et al. and measure how delaying irrevocable predictions by N observations changes accuracy for a window of T observations.

Dataset

The public Newson-Krumm Seattle benchmark contains 7,531 GPS readings, 158,167 road edges (306,182 directed arcs), and a 576-edge reference route. The nominal 1 Hz trace has four sampling gaps and spans 7,611 seconds. Longitude/latitude are projected to UTM zone 10N. Speed and heading use a causal five-reading displacement. The first 4,518 points train weights, the next 753 select T/N, and the final 2,260 form the test set (60/10/30 split). Per-point truth is an offline monotone route-distance alignment, not independent ground truth; test candidate coverage is 99.91%.

Approach and implementation

Candidate states are all directed arcs within 50 m of a GPS point, found using a spatial index. Emissions combine Gaussian point-to-road density (sigma=10 m), directed heading agreement, and a high-speed/speed-limit constraint. Transitions combine a committed-history backtrack penalty and reciprocal GPS-versus-arc-midpoint distance discrepancy. Finite-window Viterbi maximizes summed log potentials, clamps already committed states at their actual indices, and commits state t-N when reading t arrives. At EOF, the remaining N states are flushed. No road-network shortest path or connectivity check is used. Balanced L1-SVM weights are fitted using ten expanding training-only time folds (15-point gap); positive coefficients become normalized potentials and intercepts are discarded. Python, Shapely, pyproj, NumPy and scikit-learn implement the pipeline. Run python run_demo.py --T 10 --N 7 after training.



Model	T / N	Directed error	Edge error
Zero delay	10 / 0	13.89%	8.67%
Validation-selected	10 / 7	9.60%	4.60%
Nearest segment	- / 0	29.91%	8.01%

Results and conclusions

The validation-selected model runs in 0.490 s/1,000 points, excluding loading. Seven EOF predictions are flushed. Buffers improve accuracy, but minima are not consistently at T/2 and curves are not uniformly convex. Dropping distance discrepancy improves directed error to 8.94%. Proxy labels, one trip, sampling gaps and disconnected predictions limit conclusions. Future work: independent labels, more routes, turn features and causal filtering. GitHub URL is not supplied; publishing remains pending.

References

Ahres et al., supplied CS229 paper; Newson & Krumm (2009), ACM GIS, doi:10.1145/1653771.1653818; Forney (1973), The Viterbi Algorithm.